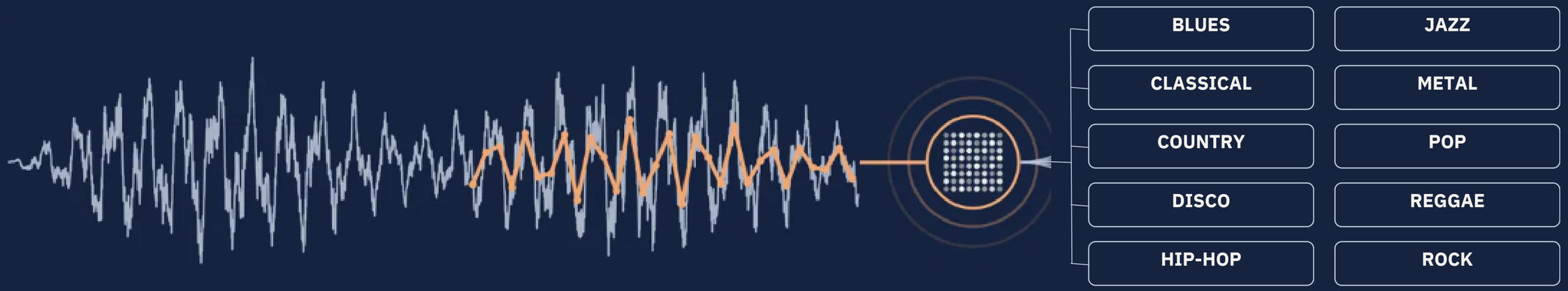
UE24CS352A MACHINE LEARNING · MINI-PROJECT

Latent Feature Extraction for Musical Genres from Raw Audio

GTZAN: 1,000 songs across all 10 genres



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github.com/Harsha-Kamaraj/ML-Mini-Project

THE PROBLEM

Can a model recognise genre from sound?

INPUT

A 0.907-second music clip.
Raw waveforms through Step 3.
Mel spectrograms in Step 4.

OUTPUT

One of 10 genre labels and
a 64-dimensional embedding
that summarises the clip.

BEST MODEL

73.4%

clip accuracy on unseen songs
Mean across five seeds

Our task Implement the paper models, improve them, and evaluate all 10 genres

DATASET · GTZAN

1,000 songs produce 20,000 short clips

1 song

About 30 seconds
at 22,050 Hz,
decoded
as waveform samples

→

20 clips

20,000 samples each
About 0.907 seconds

→

500 values

Average every 40
waveform samples

→

20,000

clips in the initial
paper-model dataset

10 genres × 100 songs

Blues, classical, country, disco, hip-hop, jazz, metal, pop, reggae and rock.

Every recording checked

All 1,000 files decode successfully. The song split keeps identical recordings in the same partition.

The paper's three models

Two-layer NN

500 inputs, 128 hidden units and 10 output classes.
The classification baseline.

Vanilla autoencoder

Compresses each clip to 64 values, then reconstructs it.
Uses no genre labels.

Deep Softmax AE

Adds a 10-class prediction head to the autoencoder's 64-dimensional embedding.

Loss: 0.9 × squared reconstruction error + 0.1 × cross-entropy

Adam at 0.0001, batch size 512, dropout rate 10%

Epochs: NN 200, vanilla AE 400, Deep Softmax AE 500

PART 1 · RESULTS ON THE FULL DATASET

Paper-model results on the ten-class task

Metric	Seed 0	Five-seed mean	Range across seeds
Two-layer NN: test	17.5%	17.1%	15.9–18.2%
Deep Softmax AE: train	73.5%	75.6%	73.5–77.5%
Deep Softmax AE: test	**28.5%**	**28.0%**	**26.2–28.8%**

A harder classification task

The paper evaluated four genres. These results cover all ten, so the scores are not directly comparable.

Measured from our own runs

The table uses the paper settings and clip split. The appendix checks seeds and alternative code settings.

PART 1 · DIAGNOSTIC CHECKS

What the ten-genre run revealed

2.5 pts

Variation across seeds

Deep Softmax AE test accuracy ranges from 26.2% to 28.8% on different clip splits.

50.4%

Alternative code settings

Mean training accuracy using the appendix's alternative loss, dropout and split settings.

0%

Collapsed decoder

Positive reconstruction outputs in the seed-0 joint model. Step 1 uses a linear output.

These checks describe the current experiment. They do not explain away the difference between four and ten classes.

PART 2 · EVALUATION

Each classifier uses two splits and five seeds

Clip split

15,000 train, 2,500 dev and 2,500 test clips.
Clips from a test song can also occur in training.
Useful for the paper-style protocol.

Song split

Targets 800 train, 100 dev and 100 test songs.
Whole songs and exact duplicates stay together.
Measures performance on unseen songs.

5 seeds

Mean and standard deviation

Dev set

Selects settings and checkpoints

Test set

Measures final performance

Six classifiers × two splits × five seeds = 60 evaluations

OUR MAIN MODEL · STEP 3

Convolutional Deep Softmax Autoencoder



Local sound patterns

Shared filters detect useful patterns throughout a clip. Four blocks lead to the compact embedding.

70.6% / 61.2%

Test clip accuracy on clip / song splits. Mean across five seeds.

Selected settings: pool size 4, reconstruction weight 0.1, Adam 0.001, batch size 64

EXTRA MODEL · STEP 4

Mel Spectrogram Deep Softmax Autoencoder

Raw sound

20,000 samples
About 0.9 seconds

→

Mel spectrogram

64 bands × 40 frames
2-D convolution

→

64-dimensional embedding

Learned from time and frequency patterns

→

Genre classifier

64 → 32 → 16 → 10

Rebuilt image

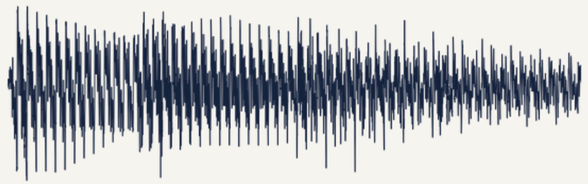
Transposed conv layers

Time and frequency information

A log-mel spectrogram makes frequency structure explicit. The model learns with 2-D filters.

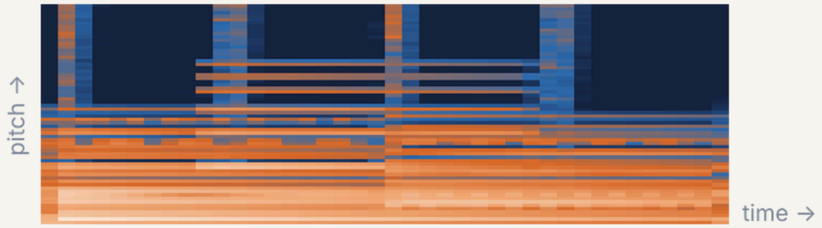
87.3% / 73.4%

Test clip accuracy on clip / song splits. Mean across five seeds.



1. Raw sound · 20,000 numbers

40 time steps ×
64 pitch bands →



2. Mel spectrogram · 64 × 40 picture

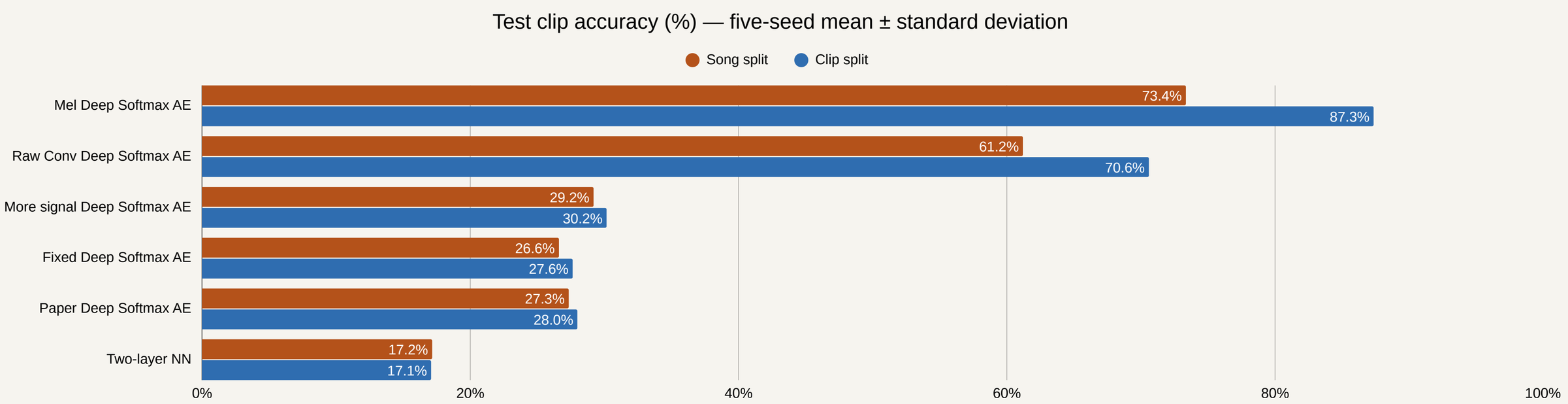
Horizontal stripes
held notes and bass

Vertical stripes
drum hits

Mel scale
more rows for low notes

RESULTS

Convolutional models lead on both splits

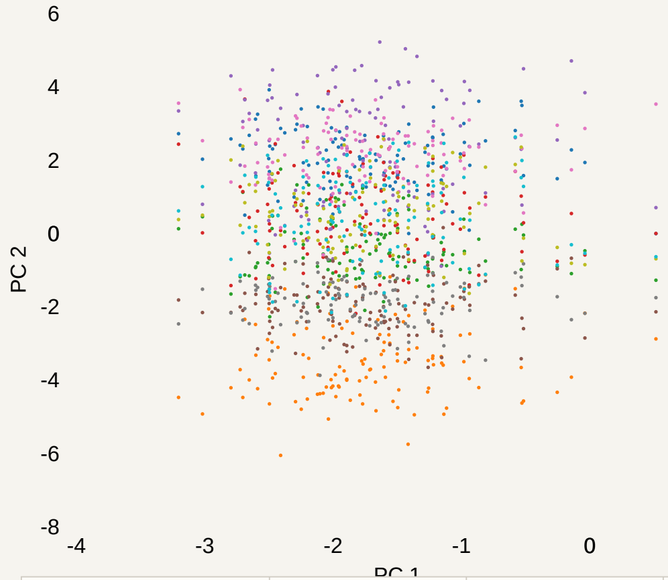


Model	Two-layer NN	Paper Deep Softmax AE	Fixed Deep Softmax AE	More signal Deep Softmax AE	Raw Conv Deep Softmax AE	Mel Deep Softmax AE
Clip split	17.1%	28.0%	27.6%	30.2%	70.6%	87.3%
Song split	17.2%	27.3%	26.6%	29.2%	61.2%	73.4%

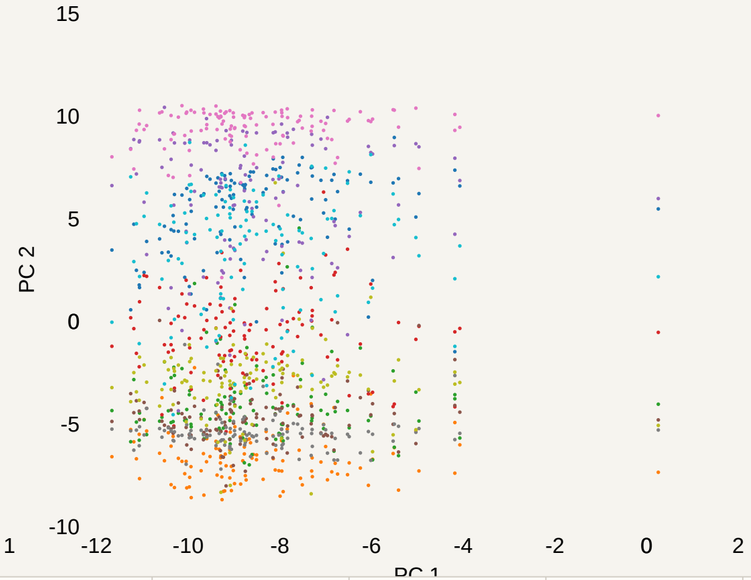
BEST MODEL · STEP 4, MEL SPECTROGRAM

Ten-genre analysis of the best model

PCA of 64-D embedding



PCA of 10-D logits



Normalised confusion matrix

True / pred.	Bl	Cl	Co	Di	Hi	Ja	Me	Po	Re	Ro
blues	0.91	0.01	0.04	0.00	0.00	0.01	0.00	0.00	0.02	0.01
classical	0.00	0.95	0.01	0.00	0.00	0.02	0.00	0.00	0.00	0.01
country	0.02	0.01	0.80	0.01	0.01	0.02	0.00	0.02	0.02	0.09
disco	0.00	0.00	0.02	0.83	0.02	0.00	0.00	0.05	0.02	0.05
hiphop	0.02	0.00	0.01	0.02	0.80	0.00	0.01	0.06	0.05	0.04
jazz	0.00	0.02	0.01	0.00	0.00	0.93	0.00	0.02	0.00	0.01
metal	0.01	0.00	0.00	0.00	0.00	0.01	0.89	0.00	0.00	0.08
pop	0.00	0.00	0.01	0.01	0.02	0.01	0.00	0.88	0.00	0.06
reggae	0.00	0.00	0.02	0.01	0.01	0.00	0.00	0.01	0.90	0.04
rock	0.02	0.00	0.05	0.02	0.00	0.01	0.05	0.01	0.01	0.84

Test F1	Blues	Classical	Country	Disco	Hip-hop	Jazz	Metal	Pop	Reggae	Rock
Mel model	0.93	0.96	0.81	0.88	0.86	0.93	0.90	0.86	0.89	0.75

Seed 0, clip split. PCA fits all 20,000 clips, with 1,200 displayed. Colours match the genre labels below.

FINAL COMPARISON

All six classifiers side by side

Model	What we changed	Train	Dev	Test
Two-layer NN	Classification baseline	28.2%	17.8%	17.1%
Paper Deep Softmax AE	Paper model with 10 outputs	75.6%	28.8%	28.0%
1 · Fixed AE	Linear decoder and training fixes	87.3%	30.1%	27.6%
2 · More signal	Extra training segments	78.5%	31.9%	30.2%
3 · Raw convolution	**1-D convolution on waveforms**	**93.0%**	**72.4%**	**70.6%**
4 · Mel spectrogram	2-D convolution on log-mel inputs	98.5%	87.8%	87.3%

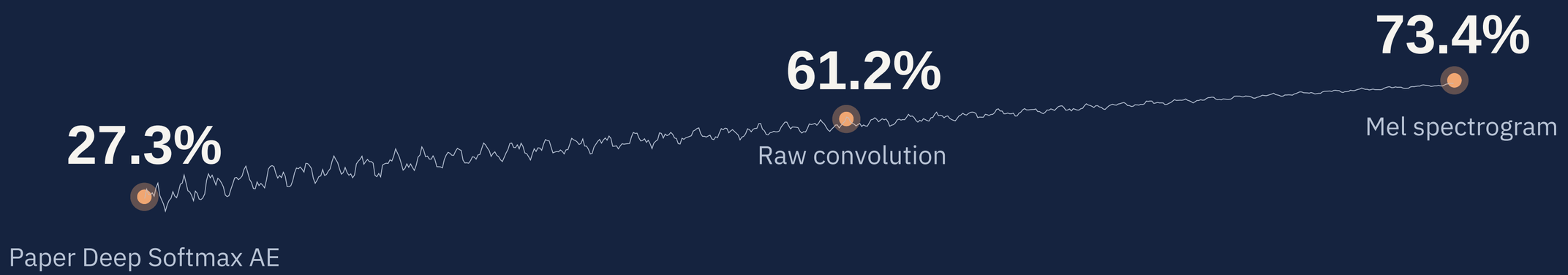
Clip-split accuracy, averaged over five seeds. The vanilla autoencoder has no classification head.

On unseen songs: raw audio 61.2%, mel spectrograms 73.4% clip accuracy

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Thank you

From raw sound to genre: 27.3% → 73.4%



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Questions?